



Hygrometry

Vapor Pressure: We know that atmospheric air contains water vapor. Pressure exerted by that water vapor is known as vapor pressure. Generally at the atmospheric temperature certain volume of atmospheric air is not saturated with water vapor i.e. Space can contain more water vapor and hence we say that the space is unsaturated with water vapor and the pressure exerted by that water vapor and pressure exerted by that water vapor is known as unsaturated vapor pressure.

A given volume of atmospheric air can be saturated with water vapor in two different ways:

- (1) By adding more and more water vapor in that space
- (2) By decreasing the temperature

It is found that amount of water vapor remaining same if we lower the temperature of that space continuously then at a particular temperature that space becomes just saturated with water vapor as indicated by that formation of water drops known as dew. This temperature at which the atmosphere becomes just saturated with the existing water vapor present in the atmosphere is known as dew point.

The unsaturated water vapor (U.V.P) at the atmospheric temperature = the saturated vapor pressure (S.V.P) at the corresponding dew point

Absolute Humidity: The amount of water vapor present per unit volume per atmospheric air is known as absolute humidity, using absolute humidity we cannot get any idea whether the atmospheric air is dry or humid.

Relative Humidity: It is defined as the ratio of the amount of water vapor present in a given volume atmospheric air at atmospheric temperature to the amount of water vapor which would saturate that given volume of atmospheric air, at the same temperature. Since this ratio is always less than one it is expressed in percentage.

$$R.H = \frac{\text{The amount of water vapor present in } V \text{ cc of atmosphere at atmospheric temperature } t^{\circ}\text{C}}{\text{The amount of water vapor which would saturate } V \text{ cc of atmosphere at the temperature } t^{\circ}\text{C}}$$

$$R.H = \frac{\text{The amount of water vapor present in 1 cc of atmospheric air at } t^{\circ}\text{C}}{\text{The amount of water vapor which would saturate 1 cc of air at } t^{\circ}\text{C}}$$

$$R.H = \frac{\text{The pressure exerted by water vapor at } t^{\circ}\text{C}}{\text{The pressure exerted by water vapor which would saturate the atmosphere at } t^{\circ}\text{C}} \times 100$$

$$R.H = \frac{\text{The U.V.P at } t^{\circ}\text{C}}{\text{The S.V.P at } t^{\circ}\text{C}} \times 100$$

$$R.H = \frac{\text{The S.V.P at the corresponding dew point}}{\text{The S.V.P at } t^{\circ}\text{C}} \times 100$$